

Games Table

Field	Type	Constraints	Description
game_id	SERIAL	PRIMARY KEY	Unique identifier for each game
creator_id	INTEGER	FOREIGN KEY -> Players(player_id)	Player who created the game
grid_size	INTEGER	NOT NULL	Size of game grid
max_players	INTEGER	NOT NULL	Maximum players allowed
status	VARCHAR(20)	CHECK(status IN ('waiting','active','finished'))	Current game state
current_turn_index	INTEGER	DEFAULT 0	Index of player whose turn it is
winner_id	INTEGER	FOREIGN KEY -> Players(player_id), NULLABLE	Player who won the game

Players Table

Field	Type	Constraints	Description
player_id	SERIAL	PRIMARY KEY	Unique identifier for each player
username	VARCHAR(50)	UNIQUE, NOT NULL	Player display name
created_at	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Account creation timestamp
games_played	INTEGER	DEFAULT 0	Total games played
wins	INTEGER	DEFAULT 0	Total wins
losses	INTEGER	DEFAULT 0	Total losses
total_shots	INTEGER	DEFAULT 0	Total shots fired
total_hits	INTEGER	DEFAULT 0	Total successful hits

Moves Table

Field	Type	Constraints	Description
move_id	SERIAL	PRIMARY KEY	Unique move identifier
game_id	INTEGER	FOREIGN KEY -> Games(game_id)	Game where move occurred
player_id	INTEGER	FOREIGN KEY -> Players(player_id)	Player who fired the shot
row	INTEGER	NOT NULL	Target row
col	INTEGER	NOT NULL	Target column
result	VARCHAR(10)	CHECK(result IN ('hit','miss'))	Result of the shot
move_timestamp	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Time move occurred

GamePlayers Table

Field	Type	Constraints	Description
game_id	INTEGER	FOREIGN KEY -> Games(game_id)	Associated game
player_id	INTEGER	FOREIGN KEY -> Players(player_id)	Participating player
turn_order	INTEGER	NOT NULL	Player's position in turn order
is_eliminated	BOOLEAN	DEFAULT FALSE	Whether the player is eliminated

Ships Table

Field	Type	Constraints	Description
ship_id	SERIAL	PRIMARY KEY	Unique ship placement record
game_id	INTEGER	FOREIGN KEY -> Games(game_id)	Game associated with ship
player_id	INTEGER	FOREIGN KEY -> Players(player_id)	Owner of the ship
row	INTEGER	NOT NULL	Ship row coordinate
col	INTEGER	NOT NULL	Ship column coordinate

Environment Configuration

Variable	Type	Description
TEST_MODE	Boolean	Enables test endpoints for deterministic grading
testPassword	String	Authentication password for test endpoints